

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
'De Novo' AMP Biosynthetic Process (GO:0	-0.0825843	4	5.674e-01	9.886e-01	PFAS:111 PAICS:7003.5 ATIC:7003.5 ADSS1:7003.5 NA NA
'De Novo' Post-Translational Protein Fold	0.13105097	15	7.895e-02	9.886e-01	DNAJB13:142 SDF2L1:644 HSPA8:799 HSPA14:923 PTGES3:5662.5 HSPA2:5662.5
2-Oxoglutarate Metabolic Process (GO:000	0.08438378	13	2.922e-01	9.886e-01	GOT1:14 OGDHL:660 GOT2:675 IDH1:5662.5 PHYH:5662.5 ADHFE1:5662.5
3'-UTR-mediated mRNA Destabilization (GO	-0.02413949	11	7.816e-01	9.886e-01	DND1:1094 ZFP36:2084 TRIM71:7003.5 RC3H1:7003.5 ZFP36L1:7003.5 ZFP36L2:7003.5
3'-UTR-mediated mRNA Stabilization (GO:0	-0.06303912	11	4.692e-01	9.886e-01	ELAVL4:1331 YBX3:1592 MYD88:2044 ZFP36:2084 TARDBP:7003.5 MAPK14:7003.5
3-Phosphoadenosine 5'-Phosphosulfate Me	0.10660456	8	2.965e-01	9.886e-01	SULT1A2:323 PAPSS2:5662.5 TPST2:5662.5 SULT1B1:5662.5 SULT1B1:5662.5 SULT1E1:5662.5
5S Class rRNA Transcription By RNA Polym	-0.23755463	5	6.584e-02	9.886e-01	GTF3C5:373 GTF3C3:578 GTF3C4:1652 GTF3C2:7003.5 GTF3C6:7003.5 NA
7-Methylguanosine mRNA Capping (GO:000945	0.26546942	2	1.935e-01	9.886e-01	RNMT:311 CMTR2:5662.5 NA NA NA NA
7-Methylguanosine mRNA Capping (GO:00063	0.26546942	2	1.935e-01	9.886e-01	RNMT:311 CMTR2:5662.5 NA NA NA NA
ADP Transport (GO:0015866)	-0.12074398	5	3.498e-01	9.886e-01	SLC25A24:813 SLC25A23:2170 SLC25A42:7003.5 SLC25A41:7003.5 SLC25A5:7003.5 NA
AMP Biosynthetic Process (GO:0006167)	-0.11099817	6	3.464e-01	9.886e-01	PFAS:111 APRT:1396 PAICS:7003.5 ATIC:7003.5 ADSS1:7003.5 NUDT2:7003.5
AMP Metabolic Process (GO:0046033)	0.04567587	9	6.352e-01	9.886e-01	AK9:324 ADS51:5662.5 NUDT2:5662.5 NTSC1A:5662.5 NTSE:5662.5 AMPD1:5662.5
ATF6-mediated Unfolded Protein Response	0.05376563	5	6.772e-01	9.886e-01	MBTPS1:5662.5 XBP1:5662.5 ATF6B:5662.5 DDI3:5662.5 CREBZF:5662.5 NA
ATP Metabolic Process (GO:0046034)	0.07835106	13	3.281e-01	9.886e-01	ATP5F1C:339 ATP5PD:423 NUDT2:5662.5 TGFBI:5662.5 VPS9D1:5662.5 ATP5PF:5662.5
ATP Metabolic Process (GO:0046034)	-0.02145636	25	7.105e-01	9.886e-01	PRKAG2:478 NADK:762 PNRF1:1028 FIGLN1:1715 MYH8:1858 CTNS:2217
ATP Synthesis Coupled Electron Transport	0.04792873	3	7.737e-01	9.886e-01	NDUFB6:717 NDUFS2:5662.5 NDUFV3:5662.5 NDUFV3:10845 NA NA NA
ATP Transport (GO:0015867)	-0.09707394	9	3.133e-01	9.886e-01	SLC25A24:813 CALHM1:893 SLC25A23:2170 SLC25A42:7003.5 SLC25A41:7003.5 SLC25A5:7003.5
AV Node Cell Action Potential (GO:008601	-0.17229959	6	1.439e-01	9.886e-01	CACNB2:335 SCNAB:1328 CACNA1G:2205 SCN10A:7003.5 SCN5A:7003.5 TRPM4:7003.5
AV Node Cell To Bundle Of His Cell Commu	0.05378616	4	7.096e-01	9.886e-01	SCN5A:5662.5 CXADR:5662.5 GJC3:5662.5 GJA5:5662.5 NA NA
Arp2/3 Complex-Mediated Actin Nucleation	-0.04453935	10	6.259e-01	9.886e-01	WASHC1:820 ARPC1A:977 ARPC5L:2053 WHAMM:2179 ARPC5:7003.5 ARPC3:7003.5
B Cell Activation (GO:0042113)	0.09136123	61	1.374e-02	9.886e-01	HDAC9:43 CD86:62 FLT3:159 CASP8:207 CD40:419 CD79A:527
B Cell Activation Involved In Immune Res	0.05942331	12	4.761e-01	9.886e-01	RNF168:18 MSH2:5662.5 GPR183:5662.5 CD40LG:5662.5 CD180:5662.5 LG4:5662.5
B Cell Differentiation (GO:0030183)	0.08932514	44	4.058e-02	9.886e-01	HDAC9:43 FLT3:159 CD79A:527 PTPRC:780 IL4:881 SLC25A5:5662.5
B Cell Proliferation (GO:0042100)	0.16028080	15	3.166e-02	9.886e-01	CD40:419 CD79A:527 MEF2C:761 PTPRC:780 IL4:881 SLC25A5:5662.5
B Cell Receptor Signaling Pathway (GO:00	0.09914482	26	8.032e-02	9.886e-01	CD79A:527 CTLA4:566 MEF2C:761 PTPRC:780 NFAM1:5662.5 MS41:5662.5
BMP Signaling Pathway (GO:0030534)	-0.01336705	42	7.646e-01	9.886e-01	FAM83G:157 INHBE:62 LEFTY1:835 DDX5:960 SMAD5:983 ACVR2A:1596
C-terminal Protein Amino Acid Modificati	-0.10195328	8	3.180e-01	9.886e-01	GPLD1:832 AGBL5:2195 AGBL4:2243 AGTPBP1:7003.5 FOLH1:7003.5 ATG16L1:7003.5
C-terminal Protein Deglutamylation (GO:0	-0.13526423	4	3.488e-01	9.886e-01	AGBL5:2195 AGBL4:2243 AGTPBP1:7003.5 FOLH1:7003.5 NA NA
C-terminal Protein Lipidation (GO:0006050	-0.10932155	3	5.120e-01	9.886e-01	GPLD1:832 ATG16L1:7003.5 ATG7:7003.5 NA NA NA
CD4-dicarboxylate Transport (GO:0015740)	0.03951597	10	6.653e-01	9.886e-01	SLC25A12:782 SLC25A13:5662.5 SLC25A2:5662.5 SLC25A18:5662.5 SLC1A3:5662.5 SLC1A1:5662.5
CD4-positive, Alpha-Beta T Cell Activati	-0.04575797	5	7.126e-01	9.886e-01	PKG7:605 STOML2:7003.5 TNFSF4:7003.5 R5AD2:7003.5 HMG81:7003.5
CD40 Signaling Pathway (GO:0023035)	0.11701448	7	2.837e-01	9.886e-01	CD86:62 CD40LG:5662.5 ITGB1:5662.5 PHB2:5662.5 TNIP2:5662.5 ITGA5:5662.5
CENP-A Containing Chromatin Assembly (GO	0.04567537	5	7.236e-01	9.886e-01	MIS18A:728 CENPI:5662.5 OIP5:5662.5 NASP:5662.5 HJURP:11122 NA
COPI Coating Of Golgi Vesicle (GO:004820	0.05374924	2	7.923e-01	9.886e-01	GBF1:5662.5 ARFGAP3:5662.5 NA NA NA NA
COPI-coated Vesicle Budding (GO:0035964)	0.05374924	2	7.923e-01	9.886e-01	GBF1:5662.5 ARFGAP3:5662.5 NA NA NA NA
COPII Vesicle Coating (GO:0048208)	-0.04364555	14	5.719e-01	9.886e-01	KLHL12:438 TRAPPC11:1323 TRAPPC12:2026 TRAPPC3:7003.5 TRAPPCC2L:7003.5 TRAPPCC4:7003.5
CRD-mediated mRNA Stabilization (GO:0070	-0.14997459	5	2.455e-01	9.886e-01	YBX1:306 IGFBP2P1:841 DHX9:7003.5 PABPC1:7003.5 HNRNPD:7003.5 NA
DNA Alkylation (GO:0006305)	0.07697969	15	3.021e-01	9.886e-01	DNMT3B:57 BAZ2A:397 MTRR:5662.5 HELLS:5662.5 ATRX:5662.5 DNMT3A:5662.5
DNA Catabolic Process (GO:0006308)	-0.04419696	17	5.282e-01	9.886e-01	DICER1:391 REXO4:645 DFFB:881 DNASEL1:390 DNASEL12:1106 ENDOG:7003.5
DNA Conformation Change (GO:0071103)	-0.04860193	9	6.137e-01	9.886e-01	HMG83:599 TOP3A:1756 TOP2B:7003.5 HMG81:7003.5 TOP2A:7003.5 TOP3B:7003.5

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	-0.146877141	100	4.038e-07	2.615e-03	COL20A1:29 FAM217B:76 MTG2:103 HELX2:139 CDH26:183 ZMYND8:197
NIKOLSKY_BREAST_CANCER_IQ21_AMPLICON	-0.237384250	27	1.972e-05	6.384e-02	ADAM15:92 PMVK:204 RORC:206 MTX1:436 TDRKH:504 ZBTB7B:522
KEGG_ALLOGRAFT_REJECTION	0.230933275	21	2.496e-04	5.388e-01	CD86:62 IFNG:93 FASLG:99 IL12B:143 HLA-DQB:232 CD80:296
PEREZ_TP53_TARGETS	-0.036632011	812	4.740e-04	6.139e-01	GPR143:4 CEMIP:6 APC2:25 KDM2B:26 UNKL:55 SMTNL2:67
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	-0.056870758	333	3.910e-04	6.139e-01	KCNH2:1 CHRN8:23 CHRNA3:45 HTR1D:78 ZBTB16:99 HTR6:125
WP_EFFECTS_OF_NITRIC_OXIDE	-0.395603951	6	7.912e-04	8.540e-01	NOS3:244 MB:354 AOX1:1500 XDH:1618 NOS1:2017 NOS2:2233
CHARAFE_BREAST_CANCER_LUMINAL_VS_MESENC	-0.053803326	319	1.018e-03	9.419e-01	EPSBL1:52 SEMA4A:104 FAM83H:117 MLPH:152 KRT19:198 CAPN8:238
SHARMA_ASTROCYTOMA_WITH_NF1_SYNDROM	-0.086348307	3	6.045e-01	9.986e-01	CLEC3B:1708 ADGRV1:7003.5 SLC1A3:7003.5 NA NA NA
SOGA_COLORECTAL_CANCER_MYC_DN	0.038979579	56	3.134e-01	9.986e-01	ACO2:74 SULT1A2:323 SLC44A1:614 SULT1A4:684 PLPPI:865 B3GN1S:994
SOGA_COLORECTAL_CANCER_MYC_UP	0.028444909	63	4.353e-01	9.986e-01	GPI:398 UMPS:507 SLC7A6:554 PLOF3K:734 DTYMK:985 PAICS:5662.5
CORONEL_RX7 DIRECT_TARGETS_UP	0.014331788	23	8.120e-01	9.986e-01	XRCCI:1905 TSPYL2:1000 ABAT:5662.5 KLF9:5662.5 TOP2B:5662.5 ATP5PF:5662.5
FOROUTAN_TGFB_EMT_UP	-0.008941825	138	7.205e-01	9.986e-01	SACS:64 MMP9:328 TUF1:1407 EML1:464 ARFGAP1:582 PLEK2:594
FOROUTAN_TGFB_EMT_DN	-0.060006315	81	6.224e-02	9.986e-01	CAVIN2:69 ERMP1:186 KRT19:198 SLC04A1:251 LAMA5:295 JAG2:319
FOROUTAN_PRODRAK_TGFB_EMT_UP	-0.017890028	132	4.788e-01	9.986e-01	SACS:64 MMP9:328 TUF1:1407 EML1:464 ARFGAP1:582 PLEK2:594
FOROUTAN_PRODRAK_TGFB_EMT_DN	-0.052924699	64	1.435e-01	9.986e-01	CAVIN2:69 ERMP1:186 KRT19:198 SLC04A1:251 LAMA5:295 JAG2:319
FOROUTAN_INTEGRATED_TGFB_EMT_UP	0.008761891	82	7.841e-01	9.986e-01	COL1A1:148 VGLL3:427 BMPR2:1518 COL6A3:646 GALTNT1:676 SLC26A2:5662.5
FOROUTAN_INTEGRATED_TGFB_EMT_DN	-0.048390018	57	2.067e-01	9.986e-01	CAVIN2:69 ERMP1:186 KRT19:198 KRT15:330 HTR10:805 BIRC3:885
GLASS_IGF2BP1_CLIP_TARGETS_KNOCKDOWN_DN	0.001230343	85	9.688e-01	9.986e-01	NCEH1:720 CD2AP:750 SRPK1:882 N4BP2:928 SHMT2:5662.5 HDLBP:5662.5
HOUNKPE_HOUSEKEEPING_GENES	0.001427951	694	8.993e-01	9.986e-01	RNF168:18 MBD1:58 NARS1:94 CCAR2:126 IP08:174 PDCD7:178
IBRAHIM_NRF1_UP	0.009276877	275	6.002e-01	9.986e-01	DNMT3B:57 MBD1:58 NCKAP1:92 NARS1:94 SCPEP1:149 FUBP1:152
IBRAHIM_NRF2_UP	0.019770601	358	2.016e-01	9.986e-01	NCKAP1:92 NARS1:94 TYW3:131 SUPV3L1:224 ZNF189:253 MRPS30:265
IBRAHIM_NRF3_UP	0.052325480	2	7.942e-01	9.986e-01	HMOX1:5662.5 RPN2:5662.5 NA NA NA NA
IBRAHIM_NRF1_DOWN	-0.031727555	94	2.885e-01	9.986e-01	TMEM25:72 KCN13:95 NPHP4:116 LG1:213 PIDD1:275 PBXIP1:569
IBRAHIM_NRF2_DOWN	-0.052893290	109	5.686e-02	9.986e-01	IQGAP3:34 CDK10:102 EGR1:1226 LG1:243 PPP1R16A:266 PIDD1:275
IBRAHIM_NRF3_DOWN	0.051267514	11	5.560e-01	9.986e-01	ARHGAP23:700 SMAP1:5662.5 NQO:5662.5 PLAT:5662.5 SEMA4C:5662.5 HECTD3:5662.5
SEAVEY_EPITHELIOID_HEMANGIOENDOTHELIO	0.040657707	67	2.503e-01	9.986e-01	C17orf58:205 TNFSF15:272 CDC42EP5:564 NCEH1:720 MYL2:758 PLCD3:5662.5
BENITEZ_GBM_PROTEASOME_INHIBITION_RESPON	0.020460999	31	6.934e-01	9.986e-01	ELP6:978 ASNS:5662.5 MLF2:5662.5 GPATCH2L:5662.5 NRC1:5662.5 SOXA:5662.5
BLANCO_MELO_SARS_COV_1_INFECTION_MCR5_CE	0.053275038	7	6.255e-01	9.986e-01	KIF14:5662.5 CENPE:5662.5 ASPM:5662.5 ESM1:5662.5 KIF20B:5662.5 BRCA2:5662.5
BLANCO_MELO_SARS_COV_1_INFECTION_MCR5_CE	0.173739275	4	2.288e-01	9.986e-01	HSD17B2:52 CYT11:446 FGFR4:5662.5 RASL12:10386 NA NA
BLANCO_MELO_MERS_COV_INFECTION_MCR5_CELL	-0.022826461	187	2.832e-01	9.986e-01	SUSD4:68 KCNH3:95 RBM15:5662.5 CDKN2AIP:344 ATP2A1:595 FHDCl:650
BLANCO_MELO_MERS_COV_INFECTION_MCR5_CELL	0.136302480	20	3.488e-02	9.986e-01	HSD17B2:52 REK18D:89 HSD11B1:428 CYT11:446 DNACJ2:581 PNMA2:681
BLANCO_MELO_INFLUENZA_A_INFECTION_A594_C	0.056309102	26	3.205e-01	9.986e-01	CXCL8:629 PLAUR:687 EREG:736 MRLP4:219 CCN2:5662.5 FGFBP1:5662.5
BLANCO_MELO_INFLUENZA_A_INFECTION_A594_C	-0.043790195	78	1.817e-01	9.986e-01	APC2:25 OPLAH:364 OVCA2:509 TONSL:694 SPATA2L:777 TBC1D16:794
BLANCO_MELO_HUMAN_PARAINFLUENZA_VIRUS_3	0.00366834	132	8.010e-01	9.986e-01	TNFSF10:36 MYPN:41 STAT2:129 OAS2:281 PLEKH4A:290 HCAR2:299
BLANCO_MELO_HUMAN_PARAINFLUENZA_VIRUS_3	0.001058642	81	9.020e-01	9.986e-01	PROD12:40 CCN5:79 SERPINA6:139 EFCAB12:225 CAMSAP3:433 CA9:467
BLANCO_MELO_RESPIRATORY_SYNCYTIAL_VIRUS_3	0.001938996	194	9.598e-01	9.986e-01	TNFSF10:36 STAT2:129 PLEKH4A:290 HCAR2:299 TNFRSF9:312 TNF:363
BLANCO_MELO_RESPIRATORY_SYNCYTIAL_VIRUS_3	-0.044303487	83	1.635e-01	9.986e-01	MCFL2:317 PAD12:348 MB:354 DACT2:430 HLA-DMB:453 SYNGRA4:534
BLANCO_MELO_COVID19_SARS_COV_2_INFECTION	-0.038275814	59	3.096e-01	9.986e-01	N4BP3:176 DNAH17:360 RASGRP2:389 SDR16C5:395 INHBE:622 BCL2A1:759
BLANCO_MELO_COVID19_SARS_COV_2_INFECTION	-0.086149760	51	3.344e-02	9.986e-01	KLHDC7A:195 VIL1:236 SLC25A10:324 DACT2:430 HLA-DMB:453 SYNGRA4:534
BLANCO_MELO_COVID19_SARS_COV_2_INFECTION	0.007984507	240	6.714e-01	9.986e-01	TNFSF10:36 HDAC9:43 STAT2:129 OAS2:281 NEDD4:300 TNFRSF9:312

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Lymphoma, T-Cell, Cutaneous	0.07515422	177	5.967e-04	9.712e-01	TNFSF10:36 HDAC9:43 IFNG:93 FASLG:99 FOXP3:105 STAT2:129
Hepatitis	0.07615196	201	2.134e-04	9.712e-01	AFP:6 AKR1D1:27 TNFSF10:36 SLC06A1:72 IFNG:93 FASLG:99
Hepatitis C	0.04794725	498	3.012e-04	9.712e-01	AFP:6 AFM:31 TNFSF10:36 HDAC9:43 SLC06A1:72 IFNG:93
Psoriasis	0.04585194	522	4.089e-04	9.712e-01	TNFSF10:36 SLC06A1:72 NCKAP1:92 IFNG:93 LORR:296 FASLG:99
Scleroderma	0.09146342	119	5.908e-04	9.712e-01	ENG:3 IFNG:93 PRSS55:98 CENPC:103 COL1A1:148 DNMT1:193
Tumor Immunity	0.0674431	88	5.533e-04	9.712e-01	AFP:6 CD86:62 IFNG:93 FOXP3:105 IL33:167 APC:258
46, XX Testicular Disorders of Sex Devel	0.05307510	3	7.502e-01	9.739e-01	NR5A1:5662.5 RSP01:5662.5 NR0B1:5662.5 NA NA NA
46, XY Disorders of Sex Development	0.05481569	12	5.109e-01	9.739e-01	SRD5A2:201 NR5A1:5662.5 LHCRG:5662.5 PAICS:5662.5 MAP3K1:5662.5 HSD17B7:5662.5
46, XY female	0.13710425	10	1.333e-01	9.739e-01	SRD5A2:201 WNT4:516 NR0B1:5662.5 HSD17B3:5662.5 DMRT2:5662.5 DMRT1:5662.5
5,10-Methyleneletrahydrofolate reductase	0.10658180	15	1.531e-01	9.739e-01	HPGD5:330 AKAP9:382 ABCB1:461 SLC01B1:510 XRCCI:1905 SERPINE1:5662.5
Abnormalities, Drug-Induced	0.05306987	2	7.949e-01	9.739e-01	CAT:5662.5 EPHX1:5662.5 NA NA NA NA
Abortion, Habitual	0.05308034	4	7.131e-01	9.739e-01	MTHFR:5662.5 F2:5662.5 F5:5662.5 VEGFA:5662.5 NA NA
Achondrogenesis, type IB (disorder)	0.05308557	5	6.810e-01	9.739e-01	SLC26A3:5662.5 SLC26A2:5662.5 SLC26A5:5662.5 SLC26A4:5662.5 DCN:5662.5 NA
Acidosis, Lactic	0.03216919	125	2.157e-01	9.739e-01	ALDH7A1:119 TRMT10C:280 G6PC:344 TNF:363 TFAM:413 NDUFA11:455
ACROMESOMELIC DYSPLASIA, MAROTEAUX TYPE	0.05306987	2	7.949e-01	9.739e-01	IGF1:5662.5 GDF5:5662.5 NA NA NA NA
Acute Myeloid Leukemia, M1	0.04840585	76	1.453e-01	9.739e-01	TNFSF10:36 FLT3:159 CD9:306 MPO:391 S100A8:392 AQP9:569
Acute myeloid leukemia, minimal differer	0.10829890	15	1.465e-01	9.739e-01	FLT3:159 MPO:391 ABCB1:461 NCR1:1715 FAS:5662.5 FANCB:5662.5
Adenocarcinoma of lung, stage IV	-0.02552503	11	7.695e-01	9.739e-01	ROS1:1143 FBLN2:2011 CD74:2287 TP